

Contestant Number: _____

Time: _____

Rank: _____



COMPUTER NETWORK TECHNOLOGY (300)

STATE – 2020

TOTAL POINTS

_____ (100 points)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

TEST TIME: 60 Minutes

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Choose the *best* answer to the following questions. Mark answers clearly if using a scantron form. Mark A if the statement is true. Mark B if the statement is false.

1. What do you use to isolate a group of computers within your organization?
 - a. WLAN
 - b. WAN
 - c. VLAN
 - d. Internet
2. When you pick up your wireless phone, your computer drops network connectivity. What could be the cause of the problem?
 - a. EMI
 - b. RFI
 - c. Network adapter failure
 - d. Cable failure
3. What is used to uniquely identify a host on a network?
 - a. MAC address
 - b. IP address
 - c. Bit pattern
 - d. Router name
4. What do you call it when electrical signals jump to another set of wires?
 - a. EMI
 - b. RFI
 - c. crosstalk
 - d. jumpitis
5. At what speed does a T3 line run?
 - a. 1.5 Mbps
 - b. 4.5 Mbps
 - c. 44.7 Mbps
 - d. 128 Mbps
6. Which layer in the OSI model do MAC addresses and switches use?
 - a. Physical
 - b. Data Link
 - c. Network
 - d. Transport
7. You have a host with the address of 180.24.45.120. To what class would this be address be assigned?
 - a. Class A
 - b. Class B
 - c. Class C
 - d. Class D

8. Which VPN technology is the most common and the easiest to set up?
 - a. PPTP
 - b. L2TP with IPSec
 - c. SSTP
 - d. CHAP

9. What technology used in firewalls keeps track of conversations so that it knows what to allow back into a network?
 - a. Stateless packet inspection
 - b. Stateful packet inspection
 - c. NAT filtering
 - d. Application-level gateway

10. For IPv4, what class is used for multicasting?
 - a. Class A
 - b. Class B
 - c. Class C
 - d. Class D

11. Which layer is the OSI model covers routing between networks?
 - a. Physical
 - b. Data Link
 - c. Network
 - d. Transport

12. What ports are defined above 49,152?
 - a. Well-known ports
 - b. Registered ports
 - c. Dynamic ports
 - d. Sliding ports

13. What speed does FDDI use?
 - a. 10 Mbps
 - b. 44 Mbps
 - c. 54 Mbps
 - d. 100 Mbps

14. What technology uses wires from the telephone company to provide broadband Internet connection?
 - a. cable
 - b. DSL
 - c. FDDI
 - d. Frame Relay

15. How can the following IPv6 address be condensed?

2009:0123:4040:0000:0000:000A:100B

- a. 2009::123:404:A:100B
 - b. 2009::123:404:A:1B
 - c. 2009:123:4040::A:100B
 - d. 2009::0123:4040::0::000A:100B
16. Which of the following is a characteristic of single-mode fiber?
- a. Single-mode fiber supports longer distances than multimode fiber
 - b. Single-mode fiber has increased bandwidth over multimode fiber
 - c. Single-mode fiber supports only a ring topology
 - d. Single-mode allows multiplexing of electrical and light signals.
17. Which of the following is **not** a fiber connector?
- a. FC
 - b. LC
 - c. RJ-45
 - d. MT-RJ
18. What type of communication is sent to a single designated host?
- a. unicast
 - b. broadcast
 - c. multicast
 - d. anycast
19. What do you use to create VLANs?
- a. router
 - b. switch
 - c. firewall
 - d. proxy server
20. What is the maximum speed supported by 802.11b?
- a. 1 Mbps
 - b. 10 Mbps
 - c. 11 Mbps
 - d. 54 Mbps
21. What topology does FDDI use?
- a. Bus
 - b. Star
 - c. Ring
 - d. mesh

22. Which layer in the OSI model is used to verify that data was delivered without error?
- Physical
 - Data Link
 - Network
 - Transport
23. What is the maximum number of hosts you can use for each subnet that belongs to a class C address?
- 128
 - 254
 - 256
 - 65,534
24. What type of device is used to detect malicious network activities and reports only those issues to the administrator?
- NIDS
 - NIPS
 - Internet content filter
 - NAT server
25. What type of firewall do you use to block access to certain websites or instant messengers?
- Internet content filter
 - NIDS
 - Circuit-level gateway
 - NAT filtering
26. What type of address is used to send a single set of packets to multiple hosts?
- Multicast address
 - Broadcast address
 - Unicast address
 - Anycast address
27. What is used to remap private addresses to public addresses and back?
- DNS
 - DHCP
 - WINS
 - NAT
28. Which port categories include inbound ports of HTTP, HTTPS, FTP, and DNS?
- Well-known ports
 - Registered ports
 - Dynamic ports
 - Private ports

29. What routing protocol is the most popular distance-vector routing algorithm used to determine the best routes within a network?
- RIP
 - OSPF
 - BGP
 - IGMP
30. What part of the network is the point where the responsibility of the administrator ends and the telecommunications provider's responsibility begins?
- Last mile
 - demarc
 - router array
 - PAD interface
31. What is used to identify a wireless network?
- Network ID
 - Frequency identifier
 - Wireless password
 - SSID
32. What is an important aspect of the Smurf attack?
- Spoofing
 - Man-in-the-middle
 - Reconnaissance
 - Packet sniffing
33. Which topology is the most redundant and the most expensive?
- star
 - ring
 - mesh
 - bus
34. _____ is the cable rack that interconnects MDF's and workstation devices
- Intermediate Distributions Frame
 - Programmable Switch
 - PIX Firewall
 - Point of Presence
35. What mechanism do wireless networks use to access the network?
- CSMA/CD
 - CSMA/CA
 - Token passing
 - polling
36. What model promises the most processing power?
- Centralized computing
 - Distributive computing
 - Switching computing
 - Dumb computing

37. Refer to the exhibit in the box. What command produced the following results?

```
? (172.16.1.255) at ff:ff:ff:ff:ff:ff on vmnet3 ifscope  
[ethernet]  
? (172.16.2.255) at ff:ff:ff:ff:ff:ff on vmnet4 ifscope  
[ethernet]  
? (172.16.3.255) at ff:ff:ff:ff:ff:ff on vmnet5 ifscope  
[ethernet]  
? (172.16.10.255) at ff:ff:ff:ff:ff:ff on vmnet6 ifscope  
[ethernet]  
? (192.168.0.1) at 90:c7:92:cc:dd:a7 on en1 ifscope [ethernet]  
? (192.168.0.4) at 0:c:29:f9:fc:3f on en1 ifscope [ethernet]  
? (192.168.0.255) at ff:ff:ff:ff:ff:ff on en1 ifscope  
[ethernet]
```

- a. netstat
 - b. arp -a
 - c. nbtstat
 - d. nslookup
38. What type of cable should be used as part of a fire code for a building?
- a. STP
 - b. PVC
 - c. Heat protected
 - d. Plenum-rated
39. What of the following is not a characteristic of 802.11n?
- a. Frame aggregation
 - b. Channel bonding
 - c. RFI protection
 - d. MIMO
40. What routing protocol is the most popular link-state protocol used within a large organization?
- a. RIP
 - b. OSPF
 - c. BGP
 - d. IGMP
41. What layer is the OSI model is responsible for logging on and off?
- a. Physical
 - b. Session
 - c. Application
 - d. Presentation

42. Which of the following is not a private address?
- a. 10.4.24.24
 - b. 172.16.36.45
 - c. 192.168.34.3
 - d. 200.23.34.120
43. What do you call a smaller network that is part of a larger network?
- a. host
 - b. broadcast domain
 - c. cluster
 - d. subnet
44. What layer in the OSI model is used to encrypt data?
- a. Physical
 - b. Session
 - c. Application
 - d. Presentation
45. What device acts as a high-speed modem for a LAN?
- a. Analog modem
 - b. Telco incoming router
 - c. Packet switcher
 - d. CSU/DSU
46. Which is the most secure encryption used in wireless networks?
- a. WEP
 - b. WPA
 - c. WPA2
 - d. 802.1x
47. What frequency band does 802.11ac use?
- a. 2.4 GHz
 - b. 4 GHz
 - c. 5 GHz
 - d. 10 GHz

48. What is the max speed for 802.11b?
- a. 300 Mbps
 - b. 54 Mbps
 - c. 100 Mbps
 - d. 11 Mbps
49. Which layer in the OSI model includes the cable and network adapters?
- a. Physical
 - b. Session
 - c. Application
 - d. Presentation
50. TCP and UDP are found at what layer of the OSI model?
- a. Application layer
 - b. Presentation layer
 - c. Transport Layer
 - d. Session Layer